Supplementary Materials: A Four-Dimension Pre-Modeling Audit Protocol for Educational Prediction Benchmarks

Yan Ma[1,3] Lizhuo Zhang[2,*]
[1] School of Education, Hunan Agricultural University, Changsha 410128, China
[2] School of Information and Intelligence, Hunan Agricultural University, Changsha 410128, China
[3] School of Foreign Studies, Changsha University of Science and Technology, Changsha 410114, China
[*] Correspondence: zhanglizhuo@hunau.edu.cn

This document contains supplementary tables and figures for the main manuscript.

S1 Supplementary Table

Table S1: Model-complexity ablation: instability ratio (I) and beat rate across linear and nonlinear models on 100 repeated 80/20 splits.

Dataset	Model	MAE (mean ± SD)	R^2 (mean)	r (mean)	I	Beat rate
MM-TBA	Linear	0.795 ± 0.089	-0.067	0.178	2.21	0.80
	Ridge	0.793 ± 0.089	-0.061	0.183	2.09	0.81
	Random Forest	0.815 ± 0.081	-0.063	0.196	3.96	0.65
	Gradient Boosting	0.856 ± 0.095	-0.227	0.129	4.60	0.45
Higher Ed	Linear	1.695 ± 0.199	0.041	0.469	0.90	0.83
	Ridge	1.686 ± 0.195	0.054	0.470	0.84	0.83
	Random Forest	1.193 ± 0.130	0.521	0.744	0.18	1.00
	Gradient Boosting	1.273 ± 0.146	0.464	0.711	0.23	1.00
xAPI-Edu	Linear	0.363 ± 0.025	0.614	0.786	0.12	1.00
	Ridge	0.357 ± 0.023	0.626	0.799	0.11	1.00
	Random Forest	0.308 ± 0.023	0.682	0.830	0.09	1.00
	Gradient Boosting	0.327 ± 0.024	0.663	0.818	0.10	1.00
Entrance Exam	Linear	0.598 ± 0.033	0.438	0.670	0.12	1.00
	Ridge	0.594 ± 0.032	0.448	0.676	0.11	1.00
	Random Forest	0.612 ± 0.036	0.372	0.628	0.14	1.00
	Gradient Boosting	0.595 ± 0.033	0.442	0.671	0.12	1.00
UCI Student	Linear	2.011 ± 0.148	0.242	0.530	0.36	1.00
	Ridge	2.010 ± 0.148	0.263	0.530	0.36	1.00
	Random Forest	2.005 ± 0.151	0.285	0.548	0.37	1.00
	Gradient Boosting	2.046 ± 0.151	0.257	0.531	0.41	1.00
Dropout	Linear	0.206 ± 0.006	0.651	0.807	0.021	1.00
	Ridge	0.206 ± 0.006	0.651	0.807	0.021	1.00
	Random Forest	0.153 ± 0.006	0.684	0.828	0.019	1.00
	Gradient Boosting	0.160 ± 0.006	0.692	0.832	0.019	1.00
OULAD	Linear	0.296 ± 0.002	0.471	0.687	0.010	1.00
	Ridge	0.296 ± 0.002	0.471	0.687	0.010	1.00
	Random Forest	0.193 ± 0.003	0.577	0.761	0.009	1.00
	Gradient Boosting	0.203 ± 0.002	0.610	0.781	0.008	1.00

S2 EDA Summary

Table S2: Exploratory data analysis summary for the seven audited datasets. All releases contain zero missing values. The datasets that fail the audit (MM-TBA, Higher Ed, UCI Student) do not show distinctive EDA red flags—such as extreme imbalance or redundant features—that would pre-emptively disqualify them; the fragility is only revealed through the four-dimension audit.

Dataset	N	Feat.	Target	Dist.	n_{grp}	Grp sz (min/mean/max)
OULAD	32,593	44	Pass=1, Fail/W=0	0.47:0.53	22	365/1482/2498
Dropout	3,630	36	Grad=1, Dropout=0	0.61:0.39	17	9/214/666
Entrance Exam	666	49	4-class ordinal	24%:32%:30%:15%	3	21/222/396
UCI Student	649	56	Grade $G3$ [0,20]	$\mu = 11.9$, $\sigma = 3.2$	2	226/324/423
xAPI-Edu	480	72	3-class ordinal	26%:44%:30%	12	19/40/95
MM-TBA	186	13	Lecture quality [0.5,7.5]	$\mu = 6.0$, $\sigma = 1.0$	0	—
Higher Ed	145	31	Grade [0,7]	$\mu = 3.5$, $\sigma = 2.2$	9	2/16/66

S3 Train-Test R^2 Gap

Table S3: Train-test R^2 gap across 100 repeated 80/20 splits using the same preprocessing pipeline as the main audit (pre-standardise, then split). The gap (train R^2 – test R^2) quantifies how much of the apparent iid signal is lost on held-out samples from the same distribution. On UCI Student, the linear-model train-test gap (0.105) is small relative to the iid-to-group-holdout collapse (0.242 to -0.097), confirming that the group-holdout failure is not attributable to ordinary overfitting. On OULAD, both gaps remain small.

Dataset	Model	Train R^2 mean	Test R^2 mean	Gap mean	Gap SD
UCI Student	Linear	0.347	0.242	0.105	0.091
	Ridge	0.347	0.242	0.105	0.091
	RF	0.898	0.276	0.622	0.081
	GBT	0.714	0.244	0.470	0.103
Higher Ed	Linear	0.550	0.041	0.509	0.299
	Ridge	0.550	0.054	0.496	0.292
	RF	0.937	0.518	0.419	0.140
	GBT	0.919	0.494	0.425	0.141
OULAD	Linear	0.282	0.268	0.014	0.024
	Ridge	0.282	0.268	0.014	0.024
	RF	0.912	0.376	0.536	0.024
	GBT	0.468	0.399	0.069	0.024

S4 Group-Holdout Uncertainty

Table S4: Across-group variation in group-holdout R^2. The SD quantifies how consistent the group-holdout performance is across held-out groups. For UCI Student (2 schools), the linear-model SD of 0.154 with a narrow range (-0.206 to 0.012) confirms uniform collapse rather than group-specific failure. For OULAD (22 cohorts), the across-cohort SD of 0.166 and full range (0.041 to 0.654) reflect positive—though variable—cross-group generalization.

Dataset	Model	Groups tested	Mean R^2	SD R^2	Min / Max R^2
UCI Student	Linear	2	-0.097	0.154	-0.206 / 0.012
	Ridge	2	-0.094	0.151	-0.201 / 0.012
	RF	2	0.018	0.154	-0.091 / 0.127
	GBT	2	-0.057	0.192	-0.193 / 0.079
Higher Ed	Linear	4	-8.79	7.37	-18.65 / -0.975
	Ridge	4	-8.58	7.24	-18.35 / -0.969
	RF	4	-13.0	16.20	-35.59 / -0.169
	GBT	4	-10.6	13.10	-29.11 / -0.597
OULAD	Linear	22	0.410	0.166	0.041 / 0.654
	Ridge	22	0.410	0.166	0.041 / 0.654
	RF	22	0.482	0.216	-0.052 / 0.747
	GBT	22	0.533	0.169	0.039 / 0.752

S5 Linear-Model Coefficient Stability

Table S5: Top-10 linear-model standardized coefficient stability across 100 repeated 80/20 splits for the three illustrative datasets. The sign-flip rate is the proportion of splits where the coefficient sign differs from the sign of the mean coefficient. On OULAD, all top-10 features show zero sign flips with low SD, indicating stable directional relationships. On UCI Student and Higher Ed, sign-flip rates remain zero for the top features but SDs are proportionally larger, reflecting greater uncertainty about effect magnitude.

Dataset	Feature	Mean coef	SD coef	Sign-flip rate
UCI Student	f5	−0.867	0.064	0.00
	f35	0.550	0.062	0.00
	f4	0.394	0.059	0.00
	f30	−0.341	0.060	0.00
	f14	0.289	0.068	0.00
	f17	0.260	0.066	0.00
	f11	−0.255	0.060	0.00
	f13	−0.251	0.068	0.00
	f37	−0.228	0.056	0.00
	f23	−0.221	0.114	0.01
Higher Ed	f30	0.986	0.120	0.00
	f28	0.769	0.104	0.00
	f1	0.732	0.120	0.00
	f0	−0.540	0.128	0.00
	f17	0.529	0.104	0.00
	f8	−0.429	0.088	0.00
	f4	0.420	0.128	0.00
	f2	0.392	0.108	0.00
	f19	−0.342	0.131	0.00
	f3	−0.298	0.113	0.00
OULAD	f3	0.159	0.003	0.00
	f8	−0.144	0.008	0.00
	f2	0.138	0.003	0.00
	f5	−0.135	0.007	0.00
	f6	−0.076	0.007	0.00
	f7	−0.059	0.005	0.00
	f24	−0.051	0.004	0.00
	f4	−0.045	0.008	0.00
	f1	−0.040	0.003	0.00
	f0	−0.035	0.003	0.00

S6 Random Forest Importance Stability

Table S6: Top-10 Random Forest feature importance stability across 100 repeated 80/20 splits. The coefficient of variation (CV = SD / mean) quantifies relative uncertainty. On OULAD, all top-10 CVs are below 0.12, indicating stable importance rankings. On UCI Student, CVs range from 0.08 to 0.21; on Higher Ed, from 0.18 to 0.36, reflecting greater split sensitivity in smaller datasets.

Dataset	Feature	Mean importance	SD importance	CV
UCI Student	f5	0.195	0.015	0.079
	f12	0.071	0.007	0.099
	f0	0.048	0.005	0.110
	f8	0.046	0.006	0.139
	f7	0.042	0.005	0.124
	f4	0.041	0.006	0.147
	f10	0.039	0.005	0.138
	f2	0.038	0.004	0.118
	f35	0.037	0.008	0.209
	f9	0.037	0.007	0.185
Higher Ed	f30	0.517	0.057	0.110
	f28	0.106	0.026	0.245
	f1	0.035	0.021	0.611
	f16	0.022	0.006	0.273
	f10	0.022	0.006	0.278
	f27	0.020	0.008	0.413
	f25	0.020	0.005	0.273
	f15	0.020	0.004	0.214
	f3	0.019	0.005	0.284
	f12	0.019	0.005	0.279
OULAD	f3	0.282	0.011	0.038
	f2	0.271	0.011	0.039
	f8	0.048	0.003	0.066
	f1	0.039	0.002	0.041
	f5	0.025	0.001	0.057
	f4	0.023	0.002	0.088
	f24	0.020	0.001	0.061
	f0	0.019	0.001	0.073
	f36	0.015	0.001	0.049
	f9	0.014	0.002	0.112

S7 Feature-Name Mapping

Table S7: Mapping from feature codes used in Supplementary Tables S5–S6 to human-readable feature names for the three datasets included in the feature-attribution stability analysis. Feature codes correspond to zero-indexed column positions in the preprocessed feature matrix after one-hot encoding.

Dataset	Code	Feature name
UCI Student	f0	age
	f1	Medu
	f2	Fedu
	f3	traveltime
	f4	studytime
	f5	failures
	f7	freetime
	f8	goout
	f9	Dalc
	f10	Walc
	f11	health
	f12	absences
	f13	school_GP
	f14	school_MS
	f15	sex_F
	f16	sex_M
	f17	address_R
	f23	Mjob_at_home
	f30	Fjob_other
	f35	reason_other
	f37	guardian_father
Higher Ed	f0	Student Age
	f1	Sex
	f2	Graduated high-school type
	f3	Scholarship type
	f4	Additional work
	f8	Transportation to the university
	f10	Mother's education
	f12	Number of sisters/brothers
	f15	Father's occupation
	f16	Weekly study hours
	f17	Reading frequency (non-scientific)
	f19	Attendance to seminars/conferences
	f25	Listening in classes
	f27	Flip-classroom
	f28	Cumulative GPA last semester (/4.00)
	f30	Course ID
OULAD	f0	num_of_prev_attempts
	f1	vle_total_clicks
	f2	vle_active_weeks
	f3	assessment_count
	f4	assessment_score_mean
	f5	assessment_score_std
	f6	age_band_0–35
	f7	age_band_35–55
	f8	age_band_55≤
	f9	age_band_nan
	f24	region_West Midlands Region
	f36	imd_band_30–40%

S8 Supplementary Figures

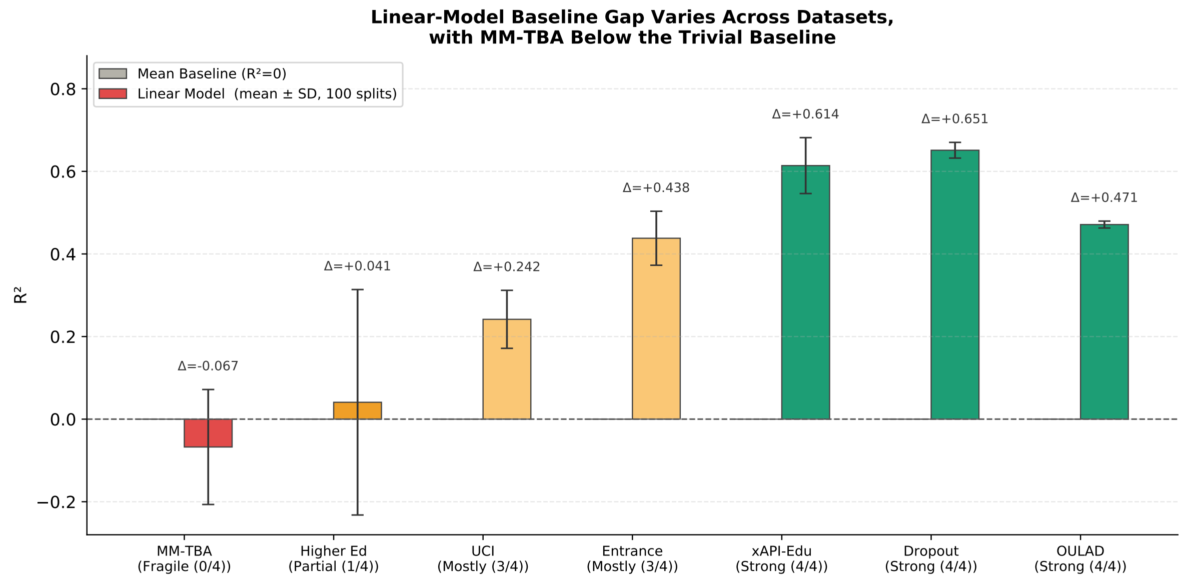

Fig S1: Baseline gap: linear-model mean R^2 (with cross-split SD error bars) over 100 random 80/20 splits for the seven datasets, ordered by readiness profile. The grey reference bar denotes the trivial mean baseline ($R^2 = 0$). The figure complements Table 2 by showing the dispersion of split-level R^2 around the reported mean; MM-TBA and Higher Ed have means near or below zero with relatively wide spread, consistent with their Fragile / Partial classifications.

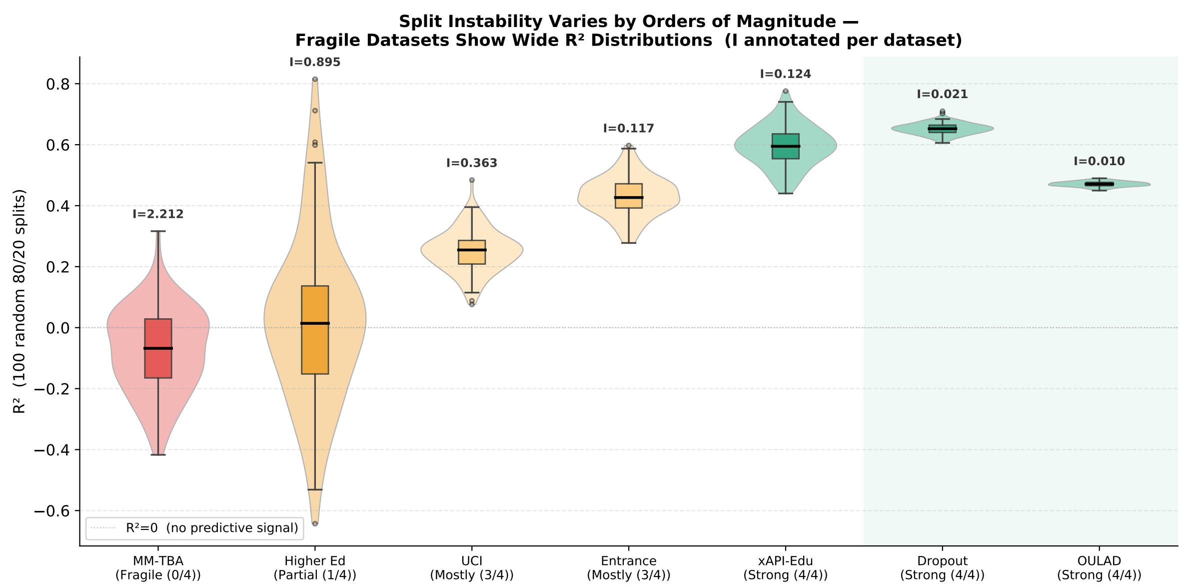

Fig S2: Split instability: violin and box plots of the linear-model R^2 across 100 random 80/20 splits for each dataset, ordered by readiness profile. The instability ratio I is annotated above each violin. Strong datasets (OULAD, Dropout) have tight distributions ($I < 0.03$); MM-TBA's distribution is broad and centred on negative R^2 ($I = 2.21$), reproducing Table 2 visually.

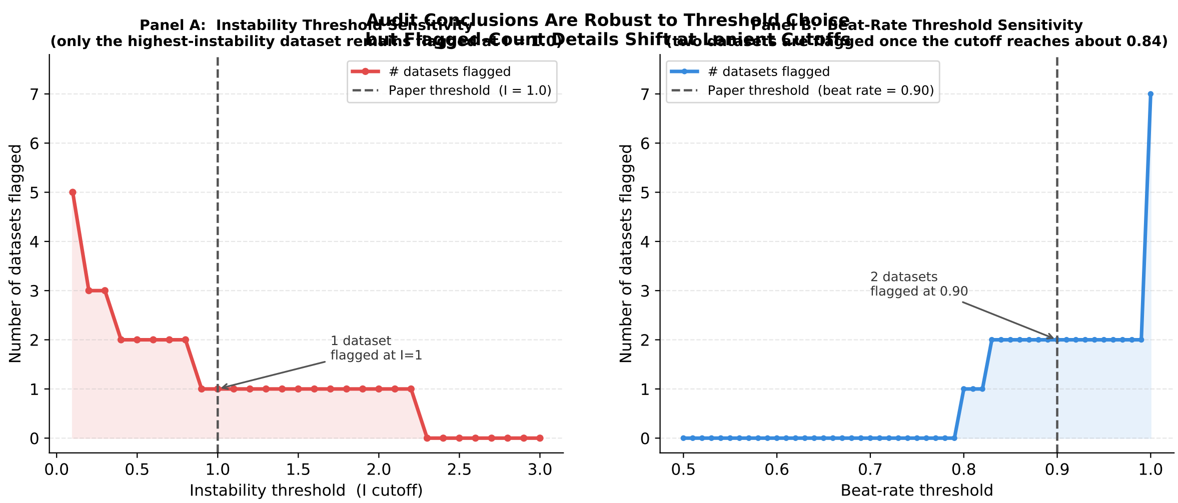

Fig S3: Threshold-sensitivity sweeps for the two operational cutoffs used in the audit. Panel (a) varies the instability cutoff over $I \in [0.1, 3.0]$ and reports the number of datasets that would be flagged as failing split instability; the paper's choice of $I = 1.0$ flags one dataset (MM-TBA, with linear $I = 2.21$), while more lenient cutoffs below about 0.9 additionally flag Higher Ed. Panel (b) varies the beat-rate cutoff over $[0.5, 1.0]$; the paper's choice of 0.90 flags two datasets, whereas more lenient cutoffs below about 0.84 reduce the flagged count. The main qualitative pattern remains unchanged: MM-TBA is always the weakest case, and Higher Ed is the only borderline dataset near the operational thresholds.